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Photochemical Control of Exciton Superradiance in Light Harvesting Nanotubes

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Abstract

Photosynthetic antennae and organic electronic materials use topological, structural and molecular control of delocalized excitons to enhance and direct energy transfer. Interactions between the transition dipoles of individual chromophore units allow for coherent delocalization across multiple molecular sites. This delocalization, for specific geometries, greatly enhances the transition dipole moment of the lowest-energy excitonic state relative to the chromophore and increases its radiative rate, a phenomenon known as superradiance. In this study, we show that ordered, self-assembled Light Harvesting Nanotubes (LHNs) display excitation-induced photobrightening and photodarkening. These changes in quantum yield arise due to changes in energetic disorder, which

in turn increases/decreases excitonic superradiance. Through a combination of experiment and modeling, we show that intense illumination induces different types of chemical change in LHNs that reproducibly alter absorption and fluorescence properties, indicating control over excitonic delocalization. We also show that changes in spectral width and shift can be sensitive measures of system dimensionality, illustrating the mixed 1-2D nature of LHN excitons. Our results demonstrate a path forward for mastery of energetic disorder in an excitonic antenna, with implications for fundamental studies of coherent energy transport.

Keywords: *excitons, superradiance, photobrightening, delocalization, coherence, light harvesting nanotubes*

Excitation induced changes in the fluorescent quantum yield are common features of solid-state optical materials. Photobrightening (PB) and photodarkening (PD) processes have been described for nanocrystal thin films (CdSe,^{1,2} PbS,³ CuBr⁴, Si⁵), organic-inorganic perovskites,⁶ and conductive organic polymers (MEH-PPV,^{7,8} P3HT⁹). While detailed mechanisms vary, most hypothesize that non-thermal charge-carrier trapping/detrapping processes underlie slow, seconds-long changes in film quantum yield due to the introduction of non-radiative decay channels.^{10,11} We observe increasing and decreasing photoluminescence quantum yield of a highly coupled molecular aggregate arising due to changes in its *radiative rate*. We attribute these changes to increased/decreased excitonic superradiance manifesting through control of energetic disorder. Unraveling this unusual mechanism gives insight into the role microscopic energetic disorder plays in determining exciton delocalization and the emissive properties of molecular aggregates. Furthermore, this work shows that different forms of aggregate disorder can be induced using excitation light, providing a path forward to understanding and using disorder modulate exciton transport in highly delocalized molecular aggregates.

In this manuscript, we describe PB and PD processes in Light Harvesting Nanotubes (LHNs). LHNs are a model system for exploring coherent exciton dynamics owing to their highly delocalized excited states, strong dipole-dipole coupling, macroscopic uniformity, and large domain sizes.^{12–17} We study LHNs in a sucrose/trehalose glass matrix, which greatly reduces photo-induced oxidative damage and maintains macroscopic order.¹⁸ We find that intense resonant illumination sequentially photobrightens and reversibly photodarkens aggregate fluorescence. Using linear and time-resolved fluorescent spectroscopy and two-dimensional electronic spectroscopy, we probe LHN lifetimes and absorption/emission lineshapes before, during, and following intense resonant excitation. We model these spectroscopic observables by introducing two different forms of energetic disorder in a model Hamiltonian, *site disorder* and *connectivity disorder*. Site disorder represents energetic variations of the site energies and/or couplings, while connectivity disorder arises from extreme local changes to a single site energy or its coupling to other sites, effectively scattering the exciton wavefunction. Through simple modeling, we show how disorder results in changes in exciton delocalization that, in turn, alter the absorption/emission peak position, lineshape, and quantum yield by changing the degree of excitonic superradiance. Observation of the recovery of aggregate superradiance points to a photochemical mechanism for photoinduced changes in disorder.

Results and discussion

Energetic disorder and excitonic superradiance. In figure 1a, we present how energetic disorder modulates excitonic dynamics in a simplified linear chain aggregate. When monomer molecular transition dipoles are assembled end-to-end their lowest energy excitonic state has a larger net transition dipole. If their transition dipole moments maintain a consistent phase relationship, these dipoles interact *coherently in phase* with an electromagnetic field leading to cooperative emission termed *excitonic superradiance*.¹⁹ In the case of parallel collinear transition dipoles, the radiative rate $k_{r,e}$ for a one dimensional J-aggregate is proportional to the effective number of coupled monomers participating to the coherent delocalization of the exciton according to equation 1.

$$k_{r,LHN} = Nk_{r,mon} \left(\frac{\lambda_{mon}}{\lambda_{LHN}} \right)^3 \quad (1)$$

Here, λ_{mon} and λ_{LHN} are the absorption maxima of exciton and monomer transition and $k_{r,mon}$ is the single monomer radiative rate.²⁰ Both static and dynamic energetic disorder decrease the coherent localization length of the exciton. Static disorder limits the coherence length by inducing buildup of random phase over N monomer units, limiting the extent of the aggregate delocalization. Dynamic disorder from environmental fluctuations further decoheres the wavefunction and leads to transient localization over a smaller number of sites. Therefore, static and dynamic disorder can act together to limit the size of the delocalized exciton and effect the radiative rate.

We use LHNs as a model system to explore extended excitonic states with superradiant emission.²¹ In figure 1b-c, we show the formation of these structures. LHNs self-assemble in water from the amphiphilic dye 3,3'-bis(2-sulfopropyl)-5,5',6,6'-tetrachloro-1,1'-dioctylbenzimidacarbocyanine (C8S3) (figure 1b). C8S3 forms tubular structures, which consist of two nested cylinders with diameters of 6 ± 1 and 13 ± 1 nm, respectively. We verify the formation of LHNs with cryo-TEM (figure 1c). In figure 1d, we show that LHN formation leads to 2250 cm^{-1} redshift of the absorption spectrum (from 19250 cm^{-1} to 17000 cm^{-1}) and the appearance of multiple narrow features. The observed redshift is the result of changes in the local environment, dipole solvation, and (primarily) significant negative coupling between monomer units, although the precise contribution of each parameter is highly dependent on details of the structural arrangement.^{12,22-24} Complementary absorption and linear dichroism spectroscopy of LHNs shows that eight features underlie the absorption spectrum. This suggest that chromophores arrange in a 2 molecule unit cell, leading to 4 electronic transitions for each wall, with 2 parallel and 2 perpendicular relative to the long axis of the linear aggregate.²⁵ This study focuses on the lowest energy band, representing the dominant parallel transition of the inner wall, from which most of the emission occurs, though some arises from the thermally occupied parallel transition of the outer wall at 16690 cm^{-1} .²⁶

In coupled aggregates, energetic disorder in each monomer unit leads to motional/exchange narrowing of the linewidth for the extended delocalized state.²⁷ In figure 1e we show two-dimensional electronic spectroscopy (2DES) on LHNS at 10K which we use to experimentally characterize the inner wall linewidth, and to study how photoinduced disorder changes aggregate lineshapes. Briefly, 2DES separates static inhomogeneous disorder along the diagonal and “dynamic” homogeneous disorder on the anti-diagonal lineshape.^{14,28} 2DES correlates the initial energy of absorption of light to the subsequent response of the system to photoexcitation. At zero population time, 2DES allows us to estimate the inhomogeneous lineshape prior to homogeneous

sources of environmental dephasing and intrinsic exciton relaxation. 2DES spectra at 10K were measured to determine the aggregate *inhomogeneous* absorptive linewidth, which will be used to model the effect of disorder LHN photophysics.

Excitation induced photobrightening and photodarkening. We describe the effect of high power photoexcitation on matrix stabilized LHNs in figure 2. In figure 2a we plot the integrated fluorescence under continuous illumination with a laser at centered at 532 nm (22.3 W/cm²). We observe an initial increase of fluorescent quantum yield (photobrightening or PB), followed by a decrease in quantum yield (photodarkening or PD) on longer timescales. This behavior is highly reproducible (figure 3a and figure S1) under both pulsed and CW excitation, and at different wavelengths of excitation (400 nm and 532 nm). We plot three sample fluorescent spectra for different irradiation times in figure 2b. We display two spectra during PB (after 3 and 50s) and the third during PD (800 s). We observe fluorescent spectral narrowing during PB and broadening during PD. Fitting to a Pseudo-Voigt profile as described in the supporting information (equation S1), in figure 2c we plot the peak position (a_1) and FWHM ($1.6371 a_2$) of the inner wall fluorescence feature derived from the modified Whiting approximation.²⁹ These two parameters show narrowing and rapid blueshift of the inner wall emission during PB, followed by broadening and sustained blueshift during PD.

In figure 2d we plot the measured radiative rate during PB and PD. We irradiate the sample with 22 W/cm² at 532 nm to induce the change in quantum yield (QY) and periodically record fluorescent lifetime data on a streak camera using a lower power 0.2 W/cm² ultrafast excitation at 400 nm, in order not to perturb the conditions established by the high power resonant excitation (shown in figure S2). The recorded lifetime data after 60 seconds of irradiation is shown in the inset of figure 2d, which we fit to a single exponential decay, $a_0 \exp(-k_{tot}t)$. The recorded total rate k_{tot} is shown in figure S2b. We calculate the radiative rate as

$$k_r(t) = k_{tot}(t)QY(t) = k_{tot}(t) \frac{QY(0)}{I(0)} I(t) \quad (2)$$

Where $QY(t)$ is extracted from the integrated fluorescence ($I(t)$) as a function of illumination rescaled by $QY(0)$ measured under low flux in an integrating sphere (supporting information for details). We observe an increase of the total rate k_{tot} and the radiative rate during PB, followed by a decrease during PD. These trends are reproduced with higher power excitation at 400 nm, consistent with the lower absorption of LHNs in that spectral region.

In figure 3a we show PB and PD behavior as a function of excitation power. We fit total fluorescent intensity to the sum of an exponential rise and an exponential decay to describe PB and PD respectively:

$$I(t) = a_0(1 - e^{-k_1 t}) + a_1 e^{-k_2 t} \quad (3)$$

We plot the rates $k_{1,2}$ in figure 3b. Within the analyzed range of excitation power, the PB rate scales linearly with the illumination power per unit area, while the PD rate shows nonlinearity at higher flux.

In figure 3c-f we examine the photodarkening behavior in more detail. In figure 3c we demonstrate that the photodarkened state slowly recovers with no excitation. We see recovery after

12 hours (3c), and partial recovery in 5 minutes (3d). Photodarkening behavior also manifests in the collected 2DES signal shown in figure 3e and SI figure S3. We observe similar behavior to that shown for fluorescence in figure 3. Decreased signal accompanies a shift in the peak location and peak FWHM (figure 3f). A partial recovery in all parameters also occurs after 90 minutes in the dark. Interestingly the recovery happens both at room temperature (figures 3c-d) and at 10K (figures 3e-f) suggesting weak temperature dependence.

The observed photobrightening and photodarkening behavior is distinct from temperature induced changes in aggregate spectra. As we show in figure S6, and in a prior publication,¹⁸ increasing temperature results in decreased intensity, broadening of spectral features and a distinct *red-shift*, while photodarkening results in a *blue-shift*. Coupling this observation with the observation of very slow recovery dynamics leads us to conclude that a distinct, non-temperature related effect is responsible for the observed photodarkening.

Modeling Static Disorder in LHNs: We apply a “rolled-brick” model of LHN morphology to understand how lineshapes and radiative rates change in response to a change in *site* static disorder. Using the inner wall radius from TEM and the assumption of closely-packed molecules with helical symmetry, we build a simple model of the inner wall LHN absorption and emission following prior methods.^{22,30} C8S3 “bricks” with x-y projection 2 x 0.4 nm (figure 4a) are tiled in a 2-D brick layer structure with a slip of 0.75 nm between successive layers of bricks. This enforces net negative coupling, and thus overall redshift of the absorption peak and J-aggregation. We then wrap the structure along a chiral vector with length set by the circumference of the LHN shown in figure 4b. The wrapping leads to splitting and redistribution of absorption cross-section of parallel- and perpendicular-polarized transitions of the LHN. To determine the coupling energy between aggregates, the geometry of a C8S3 molecule, modified to fit the PM6 basis set, and optimized with a PM6 Hamiltonian³¹ is implemented in Gaussian09³² in the gas phase (see SI and figure S7 for details). Transition density charges are extracted from standard ZINDO/S calculations based on this optimized geometry.³³ Couplings between monomers are calculated using these transition density charges.

The site-basis Hamiltonian shown in figure 4c

$$H = \sum_{n \neq m} J(n, m) a_n^\dagger a_m + \sum_n \epsilon_n a_n^\dagger a_n \quad (4)$$

Where $J(n, m)$ is the coupling between the excited states in site n and m , $a_n^\dagger$ is the raising operator for site n , and ϵ_n is a Gaussian random variable with mean ϵ_0 , the excitation frequency of an individual C8S3 molecule, and standard deviation σ . To determine the excitonic properties we diagonalize the matrix. The corresponding excitonic wavefunctions are $\phi_n = \sum_m q_{mn} \epsilon_n$, with energies d_n and corresponding overall transition dipole $\mu_n = \mu_{mon} \sum_m q_{mn} \hat{\mu}_m$ where $\hat{\mu}_m$ is the unit vector parallel to the transition dipole of molecule m .

Figure 4d shows the density of states calculated: $DOS(\omega) = \langle \sum_n \delta(\omega - d_n) \rangle$. The inhomogenous (zero-temperature) absorption spectra is dipole weighted DOS, while the emission spectra is further weighted with a Boltzmann distribution *e.g.* $PL(\omega) = \left\langle \sum_n |\mu_n|^2 \delta(\omega - d_n) \frac{1}{Z_n} e^{-\beta d_n} \right\rangle$ with $Z_n = \sum_n e^{-\beta d_n}$. This approach allows us to find the amount of site disorder that

agrees with the inhomogeneous lineshape due to static disorder. We find that $\sigma = 582 \text{ cm}^{-1}$, for the distribution of excitation frequencies gives good agreement to a 150 cm^{-1} FWHM at measured using 2DES at 10K, and low temperature absorption spectra of the parallel inner wall feature (figure S8).

We use this model as a minimal basis to explore the effect of different forms of static disorder on spectral observables. We note more sophisticated models have been developed to explain observed splittings in parallel and perpendicular spectra.^{23,24} Furthermore, recent work has highlighted possible importance of charge transfer coupling on coupling between nearest neighbours in perylene tetracarboxydiimide (PDI) based materials and conductive polymer systems.³⁴ To date, no work has been done to calculate the magnitude of CT couplings for cyanine dye aggregates, though it may provide additional insight into the observed spectral features.

Modeling Photobrightening and Photodarkening: In figure 5 we simulated the inner wall emission spectrum varying the underlying Hamiltonian to include the effects of photoinduced changes in disorder. In 5a we plot the emission spectra for three different Gaussian distributions of site disorder. We observe an increase in the intensity (5c), a decrease in the FWHM and a blueshift in the peak emission energy in figure 5d, matching the experimental results presented in figure 3c. We conclude that PB occurs due to light-induced *annealing* of the aggregate, which increases the exciton delocalization by reducing disorder in the molecular lattice. Our results suggest that starting from an initial site disorder of 500 cm^{-1} , we observe $\sim 50 \text{ cm}^{-1}$ decrease in the standard deviation in site energies. This small change in static disorder can increase the observed quantum yield by nearly 30% matching our experimental results.

During PD, we observed reversal of all trends *except* the blueshift of the peak position. PD is modeled in figures 5e-h by “removing” individual monomers from the aggregate structure for 582 cm^{-1} of static disorder (introducing connectivity disorder). We plot the emission spectra in figure 5f with different concentrations of “turned off” monomers within an aggregate. This process leads to a decrease in PL intensity (figure 5g) accompanied with an increase in the FWHM and peak energy figure 5h. Our results suggest that site deletion is a possible mechanism for photodarkening leading to emission peak shift and FWHM.

In figure 5i we explore the photo darkening behavior more fully. Here we show the scatter plot of changes in linewidth *versus* changes in peak location of the inner wall emission in the PD region. Since this type of plot reveals the connectivity and the dimensionality of the system, we also present the numerical results of idealized, nearest-neighbor-coupled cubic lattices of different dimensions. Note that both quantities have been normalized by the total amount of peak shift compared to monomers. Given small amount of site deleted, excitons in higher dimensional aggregates are rather robust against site deletion, where in 1D the connectivity between adjacent segments is completely destroyed by even a single site deleted. This justifies the treatment of the site deletion effect perturbatively, where the peak shift and width are given by^{35,36}

$$\text{shift}(\sigma) = \langle k=0 | V | k=0 \rangle = 2d\sigma \quad (5)$$

$$\text{width}(\sigma) = \pi \sum_{j \neq 0} |\langle j | V | k=0 \rangle|^2 \delta(E_j - E_0) = \pi \sigma^2 \rho_0(E_0) \quad (6)$$

where σ is the ratio of the sites deleted, $|k\rangle$ is the Bloch wavefunction with quasi-momentum k and energy E_k , V represents the excitonic interactions removed, $d=1\sim 3$ is the dimensionality, and $\rho_0(E)$

is the density of states. Both of these are given in units of the excitonic coupling. We note that the prefactor in (5) is essentially the sum of couplings of a representative site, so the direction of the shift would be opposite in an H-aggregate. The shift, given by the first order perturbation result, is a bilinear function of d and σ . On the contrary the width, given by the Golden Rule, is much more sensitive to the dimensionality.^{17,37} This explains that the higher the dimension, the steeper curve one observes as seen in the figure. On the other hand, the experimental data lies in between the results of 1D and 2D models, close to that of a more realistic rolled-brick model.

The implication of such agreement is twofold. First, this confirms that the excitons in LHN are fully delocalized over the circumference that the LHN can be regarded as a quasi-1D construct. As a consequence, the transport of excitons within enjoys both the directionality of 1D systems and the robustness against environmental disorder and noise as in 2D systems.^{17,18} Secondly, this demonstrates light-induced PD as a useful and sensitive probe to the dimensionality of similar excitonic systems. A detailed examination of disorder-induced spectral changes and their relation to dimensionality will be included in a forthcoming report. We note that the observed trends shown in figure 5i are internally calibrated and depend only on the overall dimensionality and sign of the coupling rather than details of the coupling calculation itself. This suggests that our analysis is consistent regardless of more detailed (and accurate) models of aggregate photophysics. .

The two forms of disorder needed to reproduce the spectral results point to possible chemical mechanisms for photobrightening/darkening. During excitation, the variance in site energies decreases following first order kinetics, suggesting a limited rearrangement of the aggregate structure that increases its overall site-energy order. The rate of PB varies linearly with the absorption of the aggregate, and the photon flux, which suggests that non-radiative recombination (local heating) leads to the observed reorganization of the aggregate structure. This behavior is distinct by changing the temperature of the aggregate matrix suggesting an irreversible annealing process. After the aggregate fully anneals, we begin to observe PD, with decreased PL intensity, increased linewidth and *continued peak blueshift*. PD appears to be fully reversible over hour long time-scales at both room temperature and 5K. We model this through the introduction of defect sites that do not participate in the exciton delocalization. Exciton lifetime increases during PD suggesting that these defect *do not act as non-radiative trap states*. These states instead scatter the delocalized wavefunction and decrease the amount of excitonic superradiance. We speculate a possible mechanism could be photo-induced charging of the aggregate, which has been invoked in prior experiments to explain long-lived reversible photodarkened states.¹⁰ Supporting this, saturation behavior in the rate of photodarkening observed in figure 3b is consistent with non-geminate charge recombination at high fluxes. Exploring the source of connectivity disorder will be the subject of future work.

Conclusion

LHNs have highly coherent excitons, with delocalization that extends over 10s of monomer units.²¹ Additionally, LHNs show long-range exciton motion, partially mediated by coherent dynamics.^{18,38} However, controllably modulating disorder to test the role of delocalization and the resulting coherence on energy transfer remains a significant challenge. Intense illumination leads to photobrightening and photodarkening in LHNs. For photobrightening we find an increase in aggregate quantum yield connected to a peak blueshift, a decrease in FWHM, and an increase in the

total emissive rate. Modeling this behavior suggests emission is mediated by a change in the extent of the average exciton *coherence domain* leading to changes in excitonic superradiance. Our results demonstrate that the quantum yield can increase by up to 30% with a small change in the overall static disorder.

During photodarkening we observe a decrease in quantum yield accompanied by an increased FWHM, a peak blueshift and a decreased emission rate. Interestingly, PD is reversible in the dark. This suggests a different form of disorder modulation that decreases the overall delocalization without changing the overall static disorder. We model PD by removing individual monomers from the excitonic density of states, which may reflect isomerization, individual site charging, or long-lived triplet formation. We speculate that the formation of charged monomer units can lead to the slow yet reversible PD observed. This form of PD is a sensitive independent metric of the dimensionality of the aggregate system, and points to how a quasi one-dimensional system can be robust to common forms of disordering.

Illumination allows us to anneal aggregates *and* selectively introduce defect states, providing a quantitative method to explore the role of delocalization on excitonic super-radiance. The sensitivity of quantum yield and thus excitonic superradiance to parameters may explain large variance in estimated excitonic diffusion length in the literature (ranging from <50nm to 1.7 microns).^{18,24,38} The use of photoinduced modulation can enable researchers to carefully engineer disorder and study its effect on exciton motion. As energetic disorder dephases coherent excitons, modulating it provides a path forward to understanding quantum mechanical phase in exciton transport in an exciting functional antenna material.

Experimental Methods

Sample preparation. Light harvesting nanotubes (LHNs) self-assemble from amphiphilic C8S3 dye molecules (FEW Chemicals). We prepare a stock solution of 2.9×10^{-3} M C8S3 in methanol (99.9% GC, Sigma-Aldrich). We mix 260 μ L stock solution with 1 mL deionized water. LHNs are stored in dark over 24 hours to aggregate and stabilize. To prevent oxidative photobleaching, the aqueous LHN solution is mixed at 50/50 v/v ratio with a saturated solution of 50% by weight mixture of sucrose and trehalose (Sigma Aldrich). 100 μ L of the final solution are spread on a 0.2 mm path length cuvette (Starna cells), in order to obtain a thick film. The sample is stored in vacuum (0.5 atm) for 24 hours. LHNs are thusly embedded in a photostable amorphous dry glass sugar matrix.

Photobrightening and photodarkening. Fluorescence Intensity measurements of LHNs in sugar matrix were collected using a homemade fluorimeter (figure S4). The sample is under vacuum in a ST-100 (Janis) cold-finger cryostat to prevent photo-oxidation. We used four laser systems to assess spectral dynamics as a function of incident power: 1) A supercontinuum pulsed ultrafast laser, the Koheras SuperK Extreme (NKT Photonics, 78 MHz repetition rate, Spectral dual filter output) set to 532 nm. 2) Continuous wave excitation (Thorlabs laser diode CPS532) at 532 nm. 3) 400 nm pulsed excitation, frequency doubled from femtosecond Ti:S oscillator Coherent MIRA and 4) A Melles Griot diode laser at 400 nm. For the results plotted in figure 2a, we varied the excitation power from 2.6 W cm^{-2} to 34.2 W cm^{-2} of the SuperK. The excitation beam was focused on the sample by an 8 inch lens (25 micron Gaussian focus diameter measured using a razor blade). We

collect the epifluorescence signal using a 2 inch parabolic silver mirror, which is directed into an Ocean Optics HR2000 visible spectrometer. Identical results were collected for both pulsed and CW laser sources, though the required powers were significantly higher (on the order of 500 W/cm^2) for 400nm excitation, consistent with direct absorption of the molecular aggregate. Figure 2 of S.I. shows the results obtained in case of CW excitation at 532 nm.

Emission from the above experiment was also directed into a streak camera Hamamatsu C5680 in order to collect fluorescence lifetime during irradiation. Here the femtosecond output of a Ti:S oscillator (Coherent MIRA 800 nm output, 78 MHz repetition rate, Coherent Verdi G series 532 nm pump) was doubled to excite the LHNs at 400 nm. The streak camera was set to Syncroscan mode (using the syncroscan sweep unit M5675) with a temporal resolution of 14 ps. We induce PB and PD using the SuperK laser exciting at 532 nm (22.3 W cm^{-2}) blocking the beam every 10 s, to collect the lifetime. Lifetime data were recorded at significantly lower fluxes (0.2 W cm^{-2}), previously found induced no changes in the overall fluorescent spectrum.

Absorption measurements of illumination-induced changes in LHNs in sugar matrix were collected using a Cary 5000 spectrophotometer (Agilent). The sample was illuminated using a 530 nm M530L3 LED (Thorlabs) positioned against the cryostat window. The resulting illumination was measured to be 9 mW/cm^2 across the sample area. Spectra were collected following every 10 minutes of illumination. To collect the absorption spectrum the LED was turned off for a total of 2 minutes as the spectrophotometer scanned, and then turned back on. The experiment was carried out until a total illumination time of 120 minutes was reached, after which the sample was left under vacuum in dark to recover. Following 12 hours of recovery, the absorption spectrum was collected again.

Two Dimensional Electronic Spectroscopy. 2DES spectra were collected as described in prior publications.³⁹ A non-collinear optical parametric amplifier was pumped by a Coherent Libra Ti:Sapphire regenerative amplifier laser system with 100 fs, 800 nm pulses at 10 kHz. The NOPA spectrum was centered at $17,260 \text{ cm}^{-1}$ with a full width at half maximum of $1,100 \text{ cm}^{-1}$. The laser was focused onto a transmissive diffractive optic to generate four beams in the BOXCAR geometry. The four beams were then relay imaged by common path optics into a pulse shaper comprised of a ruled reflectance grating, achromatic cylindrical lens, and 2D spatial light modulator (SLM). In the pulse shaper, each beam is spectrally dispersed onto vertically distinct regions of the 2D SLM for independent temporal shaping. After temporal shaping, the four beams exit the pulse shaper and are relay imaged to sample position. Ultrafast laser pulses were compressed to 35 fs with an average pulse energy of 45 pJ. The beams were focused to approximately $30 \mu\text{m}$ for a fluence of about $6(4) \mu\text{J/cm}^2$. For each 2D spectrum, the coherence time delay between pulses 1 and 2 was scanned from 0 to 720 fs in 5 fs steps. The signal was measured using eight-step phase cycling and spectral interferometry to recover the full electric field. Figure S5 shows the beams spectra on top of the sample absorption spectrum.

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Supporting Information Available: Including repeat measurements, 2D electronic spectra, fitting strategy and experimental setup. Further details are provided on the calculation of couplings, modeling of absorption spectra, and role of disorder in exciton delocalization. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Figure 1. a) Illustration of the delocalization of the exciton wavefunction from isolated molecules to a linear molecular aggregate. Upon aggregation, an extended transition dipole forms among coherent monomeric units, leading to an increase in the radiative rate due to larger collective

dipoles. Energetic disorder decreases the excitonic spatial extent, as well as the transition dipole. b) Chemical structure of the C8S3 monomer and 3D tubular structure of the LHNs (from Eisele et al.²⁴) c) Cryo-TEM of as-prepared LHN aggregates in sucrose-trehalose solution. d) Absorption and emission of C8S3 monomers and LHNs. e) Two-Dimensional Electronic Spectra of LHNs at 10 K reveal the ensemble inhomogeneous linewidth of solid-state supported LHNs.

Figure 2. a) Integrated fluorescence intensity of the LHNs in solid matrix as a function of the irradiation time, dotted line shows biexponential fit (equation 3). b) Three fluorescence spectra collected after 3, 50, and 800 s of irradiation (corresponding to the circles in a). Inset shows the blueshift of the inner wall emissive band during irradiation. c) FWHM (blue) and peak energy (green) as a function of irradiation time. The band narrows during photobrightening and broadens during photodarkening, while continuously blueshifting during PB and PD. d) The radiative rate from equation 2, increasing during photobrightening and decreasing during photodarkening. Inset shows raw streak camera data fit using a single exponential.

Figure 3. a) Integrated fluorescence intensity of the LHNs in sugar matrix as a function of the irradiation time (excitation at 532nm) and illumination per unit area. b) Rates of PB and PD, as a function of incident intensity c) Fluorescence intensity recorded during illumination of the sample (green) and 12 hours later (blue) curve). We observe near complete recovery of the fully photobrightened signal, followed PD. d) Fluorescence intensity during illumination showing partial recovery of signal after the excitation is turned off for 5 minutes. e) Absorption spectra of sample taken under weak LED illumination after 2 hours and 12 hours in dark (showing partial recovery) f) Difference spectra between initial and after 2hr illumination shows slight blueshift consistent with photodarkening behavior. g) 2DES taken after 30 minutes of irradiation and 90 minutes of irradiation show shift and decrease in peak intensity. h) Analysis of the rephasing signal from the 2DES spectra (figure S3) recorded at different irradiation times and after the illumination was turned off for 90 mins. We plot the signal intensity (left axis) the inner wall peak position, and inner wall FWHM (right axis).

Figure 4. a) C8S3 molecule within brick used. b) Bricks are placed in 2-D grid with set slip Δs , then wrapped according to chiral vector C , which leads to helical aggregates. c) From this geometry, off-diagonal elements are calculated and the corresponding system Hamiltonian is diagonalized to find the density of states. d) We plot the density of states, and the absorption and room temperature emission for the inner wall of the LHN.

Figure 5. a) Changing the site disorder in the Hamiltonian result in change in the emission spectra plotted in (b). The emission intensity (c), peak shift and FWHM (d) for varying amounts of diagonal (static) disorder. In (e) we remove sites and zero out their coupling to model connectivity disorder, and plot the emission spectrum in (f). Intensity (g), peak shift and FWHM changes (h) are consistent with observed trends during photodarkening. i) We plot the change in peak width vs. peak shift, both normalized by the absolute peak shift w.r.t monomer energy, and compare to Monte-Carlo simulations of idealized aggregates with 1, 2 and 3 dimensional connectivity. We also model the rolled-brick aggregate. The data (dotted lines) agrees with the LHN model and suggests that the aggregate lies somewhere between one and two-dimensional connectivity.

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